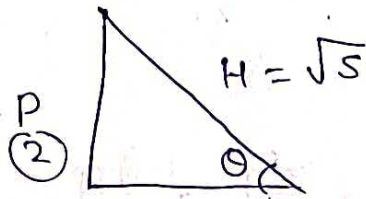


### Trigonometric #3

Question यदि  $\cos \theta = 1/\sqrt{5}$  जहाँ  $0 < \theta < \pi/2$  तो  $\frac{2 \tan \theta}{1 - \tan^2 \theta}$  के बराबर है .



$$B = 1$$

$$P^2 + B^2 = H^2$$

$$P^2 + 1 = (\sqrt{5})^2$$

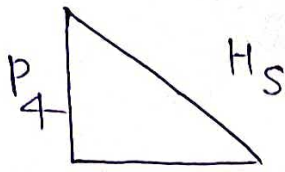
$$P^2 = 4 \rightarrow P = \sqrt{4} = 2$$

$$\frac{2 \times \left(\frac{2}{1}\right)}{1 - \left(\frac{2}{1}\right)^2} = \frac{4}{1-4} = -\frac{4}{3} \text{ Ans}$$

Question यदि  $5 \sin \theta = 4$  तो  $\frac{\sec \theta + 4 \cot \theta}{4 \tan \theta - 5 \cos \theta}$  का मान है ।

$$\frac{P}{H}$$

$$\boxed{\sin \theta = \frac{4}{5}}$$



$$B = 3$$

$$\left(\frac{S}{3}\right) + 4 \times \left(\frac{3}{4}\right)$$

$$4 \times \left(\frac{4}{3}\right) - \left(5 \times \frac{3}{4}\right)$$

$$\frac{S}{3} + 3$$

$$\frac{16}{3}$$

$$3$$

$$\left(\frac{14}{3}\right) \quad (2)$$

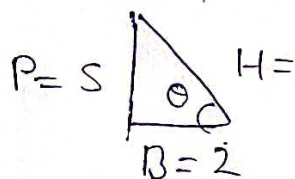
$$= 2 \underline{\underline{Am}}$$

Question यदि  $2 \sin \theta = 5 \cos \theta$ , तो  $\frac{\sin \theta + 4 \cos \theta}{\sin \theta - 5 \cos \theta}$  बराबर है।

$$2 \frac{\sin \theta}{\cos \theta} = 5$$

$$2 \tan \theta = 5$$

$$\boxed{\tan \theta = \frac{5}{2}} \quad \frac{P}{B}$$



$$5^2 + 2^2 = H^2$$

$$H = \sqrt{29}$$

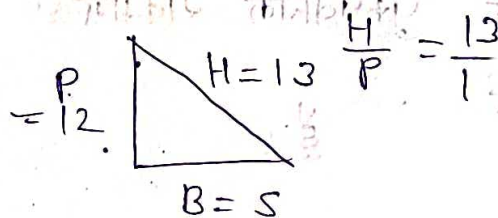
$$\frac{\cos \theta \left( \frac{\sin \theta + 1}{\cos \theta} \right)}{\cos \theta \left( \frac{\sin \theta}{\cos \theta} - 1 \right)}$$

$$\frac{\cos \theta \left( \frac{\sin \theta}{\cos \theta} - 1 \right)}{\cos \theta \left( \frac{\sin \theta + 1}{\cos \theta} \right)}$$

$$\frac{(\sin \theta + 1)}{\sin \theta - 1}$$

$$\frac{\frac{5}{2} + 1}{\frac{5}{2} - 1} \rightarrow \frac{\frac{7}{2}}{\frac{3}{2}} = \frac{7}{3} \text{ Ans}$$

Question — यदि  $\cot \theta = 13/12$  तो  $\sin \theta + \cos \theta - \tan \theta$  का मान क्या है



$$\frac{12}{13} + \frac{5}{13} - \frac{12}{5}$$

$$\frac{17}{13} - \frac{12}{5} \rightarrow \frac{85 - 156}{65} \rightarrow \frac{-71}{65} \text{ Ans}$$

Question यदि  $\sqrt{1 + \cot^2 A} = 2/\sqrt{3}$  तो  $\sin^2 A - \cos^2 A$  का मान क्या है

$$\sqrt{1 + \cot^2 A} = \frac{2}{\sqrt{3}}$$

$$1 + \cot^2 A = \frac{4}{3}$$

$$\cot^2 A = \frac{4}{3} - 1 \rightarrow \frac{1}{3}$$

$$\boxed{\cot A = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}}$$

$$A = 60^\circ$$

$$\sin^2 60^\circ - \cos^2 60^\circ$$

$$\left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

$$\frac{3}{4} - \frac{1}{4}$$

$$\frac{2}{4} = \frac{1}{2} \text{ Ans}$$

Question -  $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = \frac{4}{\sqrt{3}} \quad 0^\circ < \theta < 90^\circ$

then the value of  $(\sec \theta + \tan \theta)^{-1}$  is

$$\frac{\sin^2 \theta + (1 + \cos \theta)^2}{(1 + \cos \theta) \sin \theta} = \frac{4}{\sqrt{3}}$$

$$\frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{(1 + \cos \theta) \sin \theta}$$

$$\frac{2 + 2 \cos \theta}{(1 + \cos \theta) \sin \theta}$$

$$\frac{2(1 + \cos \theta)}{(1 + \cos \theta) \sin \theta}$$

$$\frac{2}{\sin \theta} = \frac{4}{\sqrt{3}}$$

$$\frac{2}{\sin \theta} = \frac{4}{\sqrt{3}}$$

$$\frac{2}{\sin \theta} = \frac{4}{\sqrt{3}}$$

$$\frac{1}{\sec \theta + \tan \theta}$$

$$\frac{1}{(2 + \sqrt{3})} \times \frac{2 - \sqrt{3}}{(2 - \sqrt{3})}$$

$$= \frac{2 - \sqrt{3}}{4 - 3}$$

$$= 2 - \sqrt{3}$$

$$\operatorname{Cosec} \theta = \frac{2}{\sqrt{3}} \quad \theta = 60^\circ$$

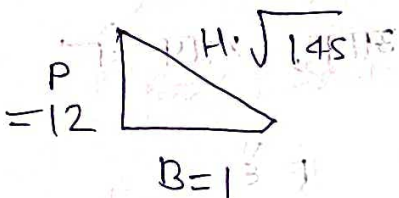
**NOTE** -  $\boxed{\sin^2 \theta + \cos^2 \theta = 1}$

Question If  $\tan \theta = \frac{12}{1}$

$$\sqrt{132 + \sqrt{132 + \sqrt{132 + \sqrt{132 + \sqrt{132 + \dots}}}}}$$

Find the value of  $\sec^2 \theta$

$$\tan \theta = \frac{12}{1} = \frac{P}{B}$$



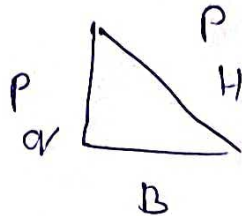
$$\sec^2 \theta = \left( \frac{\sqrt{145}}{1} \right)^2$$

$$12^2 + 1^2 = H^2$$

$$\sqrt{145} = H$$

$$= 145$$

If  $\operatorname{Cosec} \theta = p/q$   $q \neq 0$ , then find the value of  $\left( \frac{1 + \cot^2 \theta}{4 - \sec^2 \theta} \right) ?$



$$\sqrt{p^2 - q^2}$$

$$1 + \left( \frac{\sqrt{p^2 - q^2}}{q} \right)^2$$

$$4 - \left( \frac{p}{\sqrt{p^2 - q^2}} \right)^2$$

$$1 + \frac{p^2 - q^2}{q^2}$$

$$\frac{4 - \frac{p^2}{p^2 - q^2}}{p^2 - q^2}$$

$$\frac{q^2 + p^2 - q^2}{q^2}$$

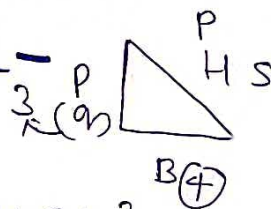
$$\frac{4p^2 - 4q^2 - p^2}{p^2 - q^2}$$

$$\frac{p^2(p^2 - q^2)}{q^2(3p^2 - 4q^2)}$$

$$\frac{p^2(p^2 - q^2)}{q^2(3p^2 - 4q^2)}$$

Ans

Second Type



$$\frac{1 + \left( \frac{4}{3} \right)^2}{4 - \left( \frac{5}{4} \right)^2}$$

$$= \frac{25}{9}$$

$$= \frac{64 - 25}{16}$$

$$= \frac{25}{39}$$

$$= \frac{25 \times 16}{39 \times 39}$$

1. If  $\sec A = \frac{17}{8}$ , while  $A < 90^\circ$ , then what is the value of the following?

$$\frac{34\sin A + 15\cot A}{68\cos A - 16\tan A}$$

यदि  $\sec A = \frac{17}{8}$  है, जबकि  $A < 90^\circ$  है, तो निम्नलिखित का मान क्या है?

$$\frac{34\sin A + 15\cot A}{68\cos A - 16\tan A}$$

(a) 23

(b) 19

(c) 30

(d) 38

2. In  $\triangle ABC$ , B is right angled, if  $\cot A = \frac{1}{2}$ , then find the value of  $\frac{\sin A (\cos C + \cos A)}{\cos C (\sin C + \sin A)}$

$\triangle ABC$  में, B समकोण हैं, यदि  $\cot A = \frac{1}{2}$  है, तो  $\frac{\sin A (\cos C + \cos A)}{\cos C (\sin C + \sin A)}$  का मान ज्ञात करें।

(a) 3

(b) -3

(c) 2

(d) -2

3. If  $\cos x = \frac{p}{q}$  and  $0^\circ < x < 90^\circ$ , then the value of  $\tan x$  is

यदि  $\cos x = \frac{p}{q}$  और  $0^\circ < x < 90^\circ$  है तो,  $\tan x$  तो का मान है।

(a)  $\frac{\sqrt{q^2 - p^2}}{q}$

(b)  $\frac{q}{\sqrt{q^2 - p^2}}$

(c)  $\frac{p}{\sqrt{q^2 - p^2}}$

(d)  $\frac{\sqrt{q^2 - p^2}}{p}$

4. If  $\cot\theta = \frac{1}{\sqrt{3}}$ ,  $0^\circ < \theta < 90^\circ$ , then find the value of  $\frac{2 - \sin^2\theta}{1 - \cos^2\theta} + \operatorname{cosec}^2\theta - \sec\theta$ .

यदि  $\cot\theta = \frac{1}{\sqrt{3}}$ ,  $0^\circ < \theta < 90^\circ$  है, तो  $\frac{2 - \sin^2\theta}{1 - \cos^2\theta} + \operatorname{cosec}^2\theta - \sec\theta$  का मान ज्ञात करें।

- (a) 1
- (b) 2
- (c) 5
- (d) 0

5. If  $\operatorname{cosec}\theta = 1.25$ , then  $\frac{4\tan\theta - 5\cos\theta + 1}{\sec\theta + 4\cot\theta - 1} = ?$

यदि  $\operatorname{cosec}\theta = 1.25$  है, तो  $\frac{4\tan\theta - 5\cos\theta + 1}{\sec\theta + 4\cot\theta - 1} = ?$

- (a) 2
- (b) 9
- (c) 1/2
- (d) 10/11

6. If  $21\tan\theta = 20$ , then  $(1 + \sin\theta + \cos\theta) : (1 - \sin\theta + \cos\theta) = ?$

यदि  $21\tan\theta = 20$ , है, तो  $(1 + \sin\theta + \cos\theta) : (1 - \sin\theta + \cos\theta) = ?$

- (a) 7:3
- (b) 5:2
- (c) 3:1
- (d) 2:1

7. If  $\tan\theta = \frac{2}{\sqrt{11}}$ ,  $0 < \theta < 90^\circ$  then  $\frac{2\operatorname{cosec}^2\theta - 3\sec^2\theta}{3\operatorname{cosec}^2\theta + 4\sec^2\theta}$  is equal to Value is.

यदि  $\tan\theta = \frac{2}{\sqrt{11}}$ ,  $0 < \theta < 90^\circ$ , तो  $\frac{2\operatorname{cosec}^2\theta - 3\sec^2\theta}{3\operatorname{cosec}^2\theta + 4\sec^2\theta}$  के समान मान है।

- (a) 11/45
- (b) 11/49
- (c) 13/49
- (d) 10/49

8. If  $\operatorname{cosec} \theta = \frac{13}{12}$ , then the value of  $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta + 9\cos\theta}$  is

यदि  $\operatorname{cosec} \theta = \frac{13}{12}$ , तो  $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta + 9\cos\theta}$  का मान है।

(a) 2

(b) 4

(b) 4

(c) 2

(d) 3

9. If  $\cot \theta = \frac{1}{\sqrt{3}}$  then the value of  $\frac{2 - \sin^2 \theta}{1 - \cos^2 \theta} + (\operatorname{cosec}^2 \theta + \sec \theta)$  is-

यदि  $\cot \theta = \frac{1}{\sqrt{3}}$  तो  $\frac{2 - \sin^2 \theta}{1 - \cos^2 \theta} + (\operatorname{cosec}^2 \theta + \sec \theta)$  का मान है-

(a) 7

(b) 4

(c) 6

(d) 5

10. If  $2 \cot \theta = 3$ , then what is the value of  $\frac{\sqrt{13}\cos\theta - 2\tan\theta}{3\tan\theta + \sqrt{13}\sin\theta}$  ?

यदि  $2 \cot \theta = 3$  है, तो  $\frac{\sqrt{13}\cos\theta - 2\tan\theta}{3\tan\theta + \sqrt{13}\sin\theta}$  का मान क्या होगा?

(a)  $\frac{1}{5}$

(b)  $\frac{2}{3}$

(c)  $\frac{3}{4}$

(d)  $\frac{1}{4}$

ANSWER SHEET

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | B | D | A | D | A | D | D | D | D  |